INCH-POUND

MIL-DTL-23661B(OS)
27 May 1997
SUPERSEDING
MIL-C-23661A(AS)
10 December 1985

DETAIL SPECIFICATION

CARTRIDGE, IMPULSE, MARK 45 MOD 0

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the Mark 45 Mod 0 Impulse Cartridge (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-D-21625 Design and Evaluation of Cartridge Actuated Devices

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8410T), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

Distribution Statement A: Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

DOD-STD-2101 Classification of Characteristics

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

DL537926 Cartridge, Impulse, Mark 45 Mod 0
DL2519737 Test Set, Closed Bomb (For Cartridge, Mark 44 and Mark 45)

PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

OP 5 Ammunition and Explosives Ashore

(Application for copies should be addressed to the Commanding Officer, Naval Publications and Forms Center, Attn: 1053, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 1742 Standard Practice for Radiographic Examination

(Application for copies of ASTMs should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Conformance to documents. The Mark 45 Mod 0 Impulse Cartridge covered by this specification shall be manufactured in accordance with DL537926.

3.3 Materials. All materials used in the manufacture of the cartridges shall conform strictly to the specifications referred to on the respective drawings. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.4 Primary components. For the purpose of this specification, the primer mix (Drawing 2406487) is considered a primary component.

3.5 Insulation resistance. Prior to installation of the bridgewire, the insulation resistance between pins and case shall not be less than 2 megohms at a relative humidity of less than 80 percent when measured with a 500-volt megohmmeter (see 4.5.1).

3.6 Bridgewire. The bridgewire shall be welded to the pins.

3.7 Bridgewire continuity. Before filling the cavity around the bridges with the igniter compound, the bridge assemblies shall be thoroughly cleaned and inspected for proper attachment and absence of projecting ends. After cleaning, the bridgewire shall be flashed by application of an electric current to assure continuity of both bridgewires. The current shall be regulated to produce a visible red glow of the bridges. An electromagnetic force (EMF) of approximately 1.5 volts ac or dc may be used with a suitable series resistor to adapt the source characteristics to the primer resistance. Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow in both bridges shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridges.

3.8 Primer mix. The primer mix shall be compounded as specified on Drawing 2406487.

3.9 Loading of igniter composition. The techniques of loading the igniter composition shall be such as to avoid damage to the bridgewire. The composition shall be smoothed off flush with the top of the cavity. All igniter composition shall be removed from exterior lateral areas of the spacer. Allow to dry thoroughly before cementing mylar disc in place. The dried composition shall be free from cracks, voids, and separation from the ignition cup. The charge shall entirely cover the bridgewire.

3.10 Propellant charge.

3.10.1 Quantity. All propellant charges shall be determined by weight rather than by volume measurement.

3.10.2 Smokeless powder: handling, and loading. The smokeless powder shall be subjected to no unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

3.10.3 Smokeless powder, condition. The smokeless powder used shall exhibit no evidence of decomposition (see 4.5.2).

3.11 Cartridge inspection.

3.11.1 Bridgewire resistance. The resistance of each bridgewire in each completely assembled cartridge, when measured electrically (see 4.5.4), shall be not less than 0.130 ohm nor greater than 0.160 ohm.

3.11.2 X-ray. Cartridges shall be free of imperfections when X-rayed (see 4.5.5).
3.12 Ballistic requirements. Cartridges shall meet the ballistic requirements specified in 3.12.1 through 3.12.3 when subjected to the test of 4.5.6.

3.12.1 Maximum pressure. The upper (U) and lower (L) limits of maximum pressure shall be U = 2300 psi and L = 1800 psi, respectively.

3.12.2 Misfire. There shall be no misfires (see 4.5.6.1).

3.12.3 Hangfire. There shall be no hangfires (see 4.5.6.2).

3.13 Workmanship. Cartridges shall be free of the following defects: burrs, dents, incomplete or damaged threads, improper crimp, perforated closure disc, bent pins, or other defects which may prevent entry of the cartridge into the firing chamber, or affect resistance of the cartridge to moisture. All cartridge marking and identification shall be verified as proper (see 4.5.3).

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.5.

4.3 First article inspection. First article inspection shall be performed after award of contract and prior to production by a facility designated in the contract or order (see 6.2). A first article sample acceptable for environmental and function testing shall be considered a sample which has met the requirements of 3.2 through 3.11.2 and 3.13. First article inspection shall be performed on sample units which have been produced using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be from the same sources of supply.

4.3.1 Sample size. The first article sample shall consist of 60 cartridges plus retain samples.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

4.3.2 Inspection routine. Sixty cartridges shall be expended in the tests listed below and a quantity of retain samples, as specified in the contract, shall be used for investigative purposes. Testing shall consist of the following tests as prescribed in MIL-D-21625: 15 “g” shock (6); vibration (6); temperature and humidity cycling (12). After environmental treatment, these cartridges are to be conditioned together with 12 untreated cartridges, at 70°F and fired as specified in 4.5.6. In addition 12 cartridges each shall be conditioned at −65°F and 160°F and fired as specified in 4.5.6 to check temperature sensitivity. Acceptance of the 70°F firings shall be in accordance with 4.5.6.

4.3.3 Failure. Failure of any cartridge shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests and examinations specified in table I.

TABLE I. Quality conformance tests and examinations.

Item	Nature of test	Requirement	Test method	Classification (DOD-STD-2101)
Body and pin assembly	Insulation resistance	3.5	4.5.1	(M101)
Cartridge	Visual inspection	3.11.3	4.5.3	(M102)
Cartridge	Bridgewire resistance	3.11.1	4.5.4	(C1)
Cartridge	X-ray	3.11.2	4.5.5	(C2)
Cartridge	Ballistics	3.12	4.5.6	(C3)
Cartridge	Preservation-packaging, packing, and marking	Section 5, Contract	4.5.3	(M103)

4.4.1 Lot size. Unless otherwise specified (see 6.2), the production lot size shall be not less than 111 and not greater than 500. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component lot may be used in more than one cartridge production lot.

4.4.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with table II.

4.4.3 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with DOD-STD-2101. Tests and examinations that verify critical characteristics are identified by the symbol (C) and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance sample size.

Lot size	Sample size	Retained sample
66 to 110	15	*
111 to 180	15	*
181 to 300	20	*
301 to 500	25	*
501 to 800	30	*
801 to 1300	35	*
1301 to 3200	40	*
3200 to 8000	50	*
8001 to 22,000	75	*

* Quantity to be specified in the contract.

4.5 Tests.

4.5.1 Insulation resistance. Prior to assembly of the bridgewire in the body and spacer assembly, the insulation resistance shall be checked with a 500-volt megohmmeter. Units with a resistance of less than 2 megohms shall be cause for rejection.

4.5.2 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.5.3 Visual examination. All cartridges shall be visually examined. Cartridges having any of the defects of 3.11.3 shall be cause for rejection. The preservation-packaging, packing, and marking of the cartridges shall be examined for conformance to section 5 and the contract (see 6.5).

4.5.4 Bridgewire resistance. Conformance with the bridgewire circuit resistance of 3.11.1 shall be ascertained on all completely assembled cartridges by means of a test circuit which limits the bridge circuit current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of current. A suitable instrument which is known to be accurate to within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which fail to meet the requirement of 3.11.1 shall be cause for rejection. The results of bridge circuit resistance measurements shall not be affected by resistance of the test circuit and connector.

4.5.5 Radiographic examination. All cartridges, in the first article and production test samples, shall be radiographically examined in accordance with ASTM E 1742. Cartridges having any observable imperfections in assembly shall be cause for rejection. The cartridges shall be positioned on their sides for the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

most revealing exposure. All cartridges shall be identified with serial numbers prior to examination. The cartridge serial numbers shall be in consecutive order beginning with number 001 in each production lot. The cartridges shall be arranged on trays or boards in consecutive numerical order , and each radiograph shall carry a permanent identification of the cartridges displayed thereon. The radiograph identification shall include the complete lot number, as stamped on the cartridges, and the span of serial numbers displayed. If any assembly or loading defects are noted in the production lot sample, the contractor shall X-ray the represented lot 100 percent and make available the X-rays to the cartridge design activity for review. Discontinuities in serial numbers shall be identified on the X-ray. Defective cartridges found by the vendor shall be clearly identified on the X-ray and those defectives removed from the production lot. The serial numbers shall be pen or ink stamped on the cartridge with approximately 1/8-inch-high characters. The serial numbers shall be located apart from the other cartridge markings.

4.5.6 Ballistic test. The number of cartridges selected from each production lot for the ballistic test shall be in accordance with table II. The cartridges shall be conditioned for 24 hours at 70 ± 5°F and fired in a closed cylindrical chamber per DL2519737. Firing current shall be not less than 7 amperes per each bridge. Lots which meet the maximum pressure requirements of 3.12.1 at an AQL of 0.10 shall be acceptable. In addition to the sample of cartridges required for the above test, additional cartridges as specified in the contract shall be furnished from the production lot for investigative purposes. Acceptance criteria shall be accept on 0 and accept on 1 regardless of lot size.

4.5.6.1 Misfire. The failure of any cartridge to fire shall be cause for rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.5.6.2 Hangfire. One or more hangfires (see 6.3.1) shall be cause for rejection of the lot represented.

4.5.6.3 Retest. There shall be no retests.

4.5.6.4 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridge, original test results should be discarded and that portion of the test reconducted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The Mark 45 Mod 0 Impulse Cartridge is used in the fuel valving system of the XXD2B target.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2).

c. Date and place of delivery.

d. Inspection conditions, if other than as specified (see 4.2).

e. Lot size, if other than as specified (see 4.4.1).

f. Sampling, if other than as specified (see 4.4.2).

g. Packaging requirements and any special marking (see 5.1, 6.5, and contract).

h. That the safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material," DOD 4145.26M are applicable. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" OP 5 are applicable.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items should be a first article sample or a sample selected from the first production items (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Hangfire. A hangfire is defined as a firing in which the time from application of current to case opening or first indication of pressure exceeds 0.075 seconds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

6.5 Contract packaging and marking. The following packing and marking requirements should be specified in the contract.

6.5.1 Packaging.

6.5.1.1 Level A. In addition to the following, Level A packaging should be in accordance with 49 CFR 171-178.

6.5.1.1.1 Inner container. Four cartridges should be packaged in accordance with DL537927.

6.5.2 Packing.

6.5.2.1 Level A. In addition to the following, Level A packing should be in accordance with 40 CFR 171-178.

6.5.2.1.1 Outer container. Twelve inner containers, as described in 6.5.1.1.1, should be packed in accordance with DL537928.

6.5.3 Marking. Cartridge marking should be as specified on Drawing 2240773. In addition, packaging and packing markings should be in accordance with 49 CFR 171-178 and MIL-STD-129.

6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

Preparing activity:
Navy - OS
(Project 1377-0058)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-23661B(OS)	**2. DOCUMENT DATE (YYMMDD)** 970527

3. DOCUMENT TITLE CARTRIDGE, IMPULSE, MARK 45 MOD O

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, INDIAN HEAD DIVISION	b. TELEPHONE *Include Area Code)* (1) Commercial (301) 743-4700 (2) AUTOVON 354-4700
c. ADDRESS *(Include Zip Code)* NAVAL SURFACE WARFARE CENTER (CODE 841OT) 101 STRAUSS AVENUE INDIAN HEAD, MD 20640-5035	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:34Z
Check the source to verify that this is the current version before use.